

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

03rd November 2015

Subject SA6 – Investment

Time allowed: Three hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the Candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** You are the Risk Manager for a life insurance company with various lines of business like participating, non-participating, unit linked and group variable insurance.

The Company is a mid-sized life insurance company with capital of Rs 1000 crores.

You are reviewing risk management models for assessing current risks and using it to form portfolio strategies.

You also need to devise a method of conducting Asset Liability Management which should highlight the risks of business appropriately.

- i)** What are the characteristics of a good risk model focussing on asset liability management and stress tests? Please state the relevant insights/information you would like to obtain from the model, which can be shared with the company's senior management as part of the management information reports. (5)
- ii)** How does the life insurance company ALM risk model differ from those of banks or general insurance companies? (5)
- iii) a)** Explain the key objectives of an investment policy (2)
- b)** Please highlight the steps involved in the process of conducting asset liability management tests? (3)
- iv)** Traditionally the company has been investing in fixed income securities and money market. Due to the drop in the bond yields the company is not getting good investment assets on the fixed income side. Meanwhile the equity markets have also witnessed a fall and the Company wants to evaluate exposure to equities.

Please give your views on suitability of equity investment for a Life insurance company (5)
- v)** How will you ascertain an optimal allocation to equities (3)
- vi)** Please discuss if addition of equity would have any impact on recently devised asset liability management framework (2)
- vii)** The duration of liability is higher than the duration of assets available under the portfolio. One of the alternatives that have been discussed is to use derivatives to reduce the duration risk since derivative could be done at a relatively less cost.

Assuming there are no regulatory restriction on the investment in derivatives, describe the various derivative instruments available in India and discuss whether they would be suitable for reducing the duration risk (10)
- viii)** Assuming the life insurance company is already listed through an IPO (Initial Public Offer) but requires further money to be raised for business expansion. For this purpose the company is evaluating the FPO, i.e. Follow-up Public Offer route instead of the OFS, i.e. Offer for Sale route.
 - a)** Please explain the difference in risk and return characteristics of money raised through an IPO and an FPO (3)
 - b)** Also explain the difference between an FPO and OFS and suggest which may be more preferable in this context? (5)

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- Q. 2)** You are an actuary consulting a regulated defined benefit (DB) pension plan of a large Indian manufacturing company called Pro. The employers at Pro are finding it difficult to fund the DB scheme. They have asked you and the trustees to explore the possibility of moving from the DB setup to a DC (defined contribution) setup. Looking at other employers, they are of the opinion that with relatively lower interest rates and worsening annuity rates, DB schemes are becoming recklessly dangerous to fund and therefore DC schemes seem more realistic. The company's pension plan manages the funds till the employees' retirement and thereafter it manages the annuity payouts through a life insurance company in India. The insurance company takes a lump sum from the pension plan of the manufacturing company and in turn provides annuities to the retiring employees under this scheme. The insurer/annuity provider considers longevity increase as a major risk. They are considering entering into a longevity swap to hedge longevity risk.

Longevity swap is a type of, longevity risk, hedging contract. Under this, the annuity provider agrees to provide a counterparty with regular pre-determined, or "fixed", payments based on agreed upon mortality assumptions. In return, the counterparty provides the pension plan with regular floating payments based on the pension plan's actual mortality experience. As a result, the pension plan has more predictable outflows over the period of the longevity swap. The counterparty to the contract assumes the longevity risk over the period covered by the contract.

- i)** Analyse the current outlook in the Indian pensions market in terms of:

- a)** Opportunities (3)
- b)** Regulatory challenges (3)
- c)** Other challenges & risks (3)

You feel that a pure DB and a pure DC are two ends of a spectrum and there a number of 'hybrid' scheme solutions where the investment and other risks can be shared by the employer and the employee.

- ii)** Compare the following schemes.

- a)** Final salary DB schemes with no employee contributions (or fixed employee contribution rate) (2)
- b)** Pure DC scheme (2)
- c)** Hybrid of both DB and DC - this would involve a final salary benefit with money purchase underpin (2)

- iii)** Please explain the longevity risk for the insurance company providing annuities to employees in the DB scheme? What can you do to deal with the longevity risk? (3)
- iv)** Highlight the risks for the annuity provider if it were to enter into a longevity swap and how such risks can be mitigated? (4)

- v) In addition to the various risks involved in longevity swap, what other important aspects the annuity provider needs to consider while entering into the swap arrangement? (5)

As mentioned above the assets in the accumulation stage are managed by the pension plan and are transferred to annuity provider, later, on retirement. The pension plan assets are partly invested in equities. The trustees have sought your help in hedging the equity exposure of the underlying portfolio by buying put options to maintain the funding level. You are analysing 2 approaches:

- a) To adopt a policy of permanently hedging the equity exposure, always rolling maturing hedges into new ones
- b) To adopt an ad hoc approach whereby you will only hedge the exposure when you think there is a material risk and you will not be able to maintain funding levels above the required level.
- vi) Compare the two alternative approaches and make a recommendation, with reasons, as to which one should be adopted. (6)
- vii) Assume that European put options will be used to hedge the equity portion. Discuss the relative merits of using short-dated (three-month) rather than long dated (nine-month) at-the-money options (4)
- viii) Discuss the use of single stock options rather than index options to hedge the equity portfolio. (4)
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- Q. 3)** i) Increasing the Repo rate, cash reserve requirement and open market operations are used as methods/tools to implement Contractionary Monetary policy.

Please explain what is the objective of a Contractionary Monetary Policy and how the implementation of each method above helps in achieving the desired objective? (4)

- ii) Recently a news which has caught everyone's attention is the anticipated hike in the Fed rates. However this has led to a concern amongst the emerging economies as any increase in Fed rates (the rate at which a depository institution lends overnight funds to another depository institution) is likely to result in outflows.

Please discuss the likely impact of hike in Fed rates on the following:

- a) Indian Currency (3)
- b) Stock Markets (5)
- c) Carry trade in the context of Indian Economy. (Carry trade refers to borrowing in a low interest rate currency and investing in risk assets such as emerging market equities and debt) (4)

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